

Joongmin Shin

Senior Researcher / Research Scientist, Korea University · Founder & Lead, KUDoc

Seoul, Republic of Korea · Open to relocation (US/EU) and remote roles

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Professional Summary

AI researcher with 5+ years of experience in multimodal LLM/LVLMs, structure-aware parsing, and retrieval-augmented generation for document-heavy settings. My current work focuses on structured evidence and reliable long-context reasoning over heterogeneous unstructured inputs, with first-author papers at NeurIPS, ACL, CVPR, and EMNLP. In addition to my individual research, I founded and lead KUDoc, the Document AI research group at Korea University, where I help shape collaborative directions across document understanding, multimodal RAG, evaluation, and applied AI. I aim to pursue a Ph.D. to extend this foundation toward structured memory, world models, and planning for future agents.

Key Strengths

- **Parsing from sentence to document level:** Build neural dependency parsers for text and extend them to **document-level trees and graphs**, enabling consistent structure-aware chunking and retrieval for long, noisy PDFs.
- **Explainable hybrid architectures:** Design and implement systems that integrate symbolic rules and knowledge graphs with LLM/LVLM models, improving both performance and transparency.
- **Domain problem definition & Rapid prototyping:** Translate complex industrial requirements into structured tasks for RAG systems, managing cross-functional teams to deliver end-to-end prototypes and industry pilots.

Education

Pusan National University

Busan, Republic of Korea

M.S. IN ARTIFICIAL INTELLIGENCE

Mar 2021 – Feb 2023

- Thesis focused on parsing unstructured data and integrating syntactic/morphological analysis for Korean NLP; participated in national AI projects and filed related patents.

Pusan National University

Busan, Republic of Korea

B.S. IN COMPUTER SCIENCE AND ENGINEERING

Mar 2016 – Feb 2021

- Completed coursework in software engineering and data analysis; engaged in multiple industry projects and hackathons, building teamwork and leadership skills.

Selected Publications

* First author

ADMIT: Support-Gated Memory-Write Admission for Document QA Agents

Joongmin Shin*, Gyuho Shim, Hyeonseok Moon, Jaehyung Seo

NeurIPS 2026 - Conference on Neural Information Processing Systems: Main

PILAR: A Page-Grounded Unified Evidence Representation via an Entity-Linked Assertion Graph for Open-Domain QA Agents over Multimodal Document Corpora

Joongmin Shin*, Gyuho Shim, Jung-hun Lee, Jaehyung Seo

EMNLP 2026 - Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing: Findings

HiKEY: Hierarchical Multimodal Retrieval for Open-Domain Document Question Answering

Joongmin Shin*, Gyuho Shim, Jeongbae Park, Jaehyung Seo, Heuseok Lim

ACL 2026 - Annual Meeting of the Association for Computational Linguistics: Main (Oral)

M3DocDep: Multi-modal, Multi-page, Multi-document Dependency Chunking with Large Vision-Language Models

Joongmin Shin*, Jeongbae Park, Jaehyung Seo, Heuseok Lim

CVPR 2026 - Computer Vision and Pattern Recognition Conference: Main

MultiDocFusion: Hierarchical and Multimodal Chunking Pipeline for Enhanced RAG on Long Industrial Documents

Joongmin Shin*, Chanjun Park, Jeongbae Park, Jaehyung Seo, Heuseok Lim

EMNLP 2025 - Proceedings of the Conference on Empirical Methods in Natural Language Processing: Main

Intelligent Predictive Maintenance RAG framework for Power Plants: Enhancing QA with StyledFS and Domain Specific Instruction Tuning

Seongtae Hong*, Joongmin Shin*, Jaehyung Seo, Taemin Lee, Jeongbae Park, Heuseok Lim

EMNLP 2024 - Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track

Professional Experience

Human-Inspired AI Research, Korea University

Seoul, Republic of Korea

Senior Researcher / Research Scientist

Jun 2023 – Present

- Evidence Graphs & Agent Memory (NeurIPS 2026 / EMNLP 2026):** Developed *PILAR*, a page-grounded evidence representation that links assertions from text, tables, and figures through an entity-linked graph for open-domain QA agents, and *ADMIT*, a support-gated admission policy that controls what document QA agents commit to memory.
- Hierarchical Multimodal Retrieval (ACL 2026, Oral):** Developed *HiKEY*, which combines global routing, local ranking, and structured evidence assembly for open-domain document question answering.
- Multimodal Document Parsing & Chunking (CVPR 2026 / EMNLP 2025):** Designed an LVLM-based document parsing framework (*M3DocDep*) for noisy, large-scale industrial PDFs, building pipelines that produce document-level dependency trees. Developed *MultiDocFusion*, a structure-aware multimodal chunking methodology that preserves hierarchical layouts to improve RAG retrieval quality.
- Domain-Specific LLM & RAG (EMNLP 2024):** Designed and implemented *StyleDFS*, a specialized Korean LLM and RAG pipeline for scientific literature with tailored preprocessing algorithms, completing technology transfer twice.
- Industrial Pilots & Prototyping:** Led multimodal RAG projects and MVPs for Samsung Fire & Marine Insurance (reducing manual search time by 40%), Hyundai Mobis (GPT-based PLC Programming Assistant), and Synapse (Multilingual Medical Consultation), successfully securing follow-up funding.
- Hybrid QA Systems:** Architected QA systems combining knowledge graphs and LLMs, resulting in a 10% accuracy improvement and a patent on adaptive rule-based filtering.

AI Lab, Pusan National University

Busan, Republic of Korea

Researcher (M.S.)

Sep 2020 – Feb 2023

- Exobrain WiseQA National Project:** Led core research on **Korean text dependency parsing**, developing neural-symbolic parsers that integrate morphological and syntactic features as the backbone for sentence- and table-level QA.
- Table QA Frameworks:** Designed a pipeline converting semi-structured financial/encyclopedia tables into structured representations, achieving state-of-the-art Korean Table QA performance and improving model explainability.
- Applied AI Collaborations:** Collaborated with KT AI2XL on Korean TTS prosody prediction using syntactic parsing features; contributed to finance QA, sentiment-based marketing, and VR recommendation projects, earning an **Outstanding Paper Award**.

Honors & Awards

2024	Deputy Prime Minister and Minister of Education Award , Competition for Best Practices in Academia-Industry Cooperation	National Research Foundation (NRF)
2023	Outstanding Paper Presentation Award , Korean Institute of Information Scientists and Engineers (KIISE)	KIISE
2023	Grand Prize , Samsung Fire & Marine Insurance LLM-based MVP Model Idea Competition	Samsung
2023	Excellence Award (Technology Track) , KU-Grant Program	Korea University
2022	Outstanding Paper Presentation Award , Korea Software Congress (KSC)	KSC
2022	Excellence Award , Student-led Research Project, Graduate School of AI	Pusan National University
2020	Milestone Scholarship (Grade A) , SW-centered University Milestone Scholarship Program	Pusan National University

Technical Skills

Programming	Python, C/C++, Java, C#, Django, React, Unity
Machine Learning	LLM/LVLM fine tuning, hybrid rule-neural models, knowledge graphs
Tools	PyTorch, TensorFlow, NumPy, Pandas, Scikit-Learn, LangChain, SQL, Linux
Project Management	Team leadership, cross-functional collaboration, proposal writing
Languages	Korean (native), English (fluent reading/writing)

Teaching & Mentoring

Pusan National University

Busan, Republic of Korea

Graduate Research Mentor

2021 – 2022

- Mentored undergraduate capstone projects on Korean dependency parsing and robust Table QA; guided students to achieve near state-of-the-art performance.
- Assisted in courses on web application programming and mobile app development, providing feedback and supporting team projects.